Neva Krien

AI developer and systems programmer with experience in AI research, technical education, Rust, and low-level development.

nevo.krien@gmail.com | +972 54-244-9864 | GitHub | Codeberg

Experience

- Mentee of Guy Tamir (Intel) since 2022; contributed to research and educational outreach in AI.
- Participated in research on LLMs for C++ code generation. See the team's first paper.
- Built demo applications for Intel products and helped deliver Intel's hands-on AI PC workshop at GenML 2024.
- Currently co-developing an accredited Practical AI course for Bar-Ilan University with Guy Tamir.
- Authored example code used in Intel's official YouTube AI tutorials.
- Contribute technical guidance and code to programming and AI livestreams hosted by PubNub CTO Stephen Blum.
- Core team member of PAL: A new programming language currently under closed development.
- Built two toy compilers/interpreters:
  - One in **pure C99 + NASM**, including a full optimization pipeline (constant folding, loop unrolling, branch elimination).
  - One in **Rust**, using `unsafe` for a simple dynamic language with a VM.
- Collaborate on API design and implementation for the Rust diagnostics library Ariadne, across its GitHub and Codeberg development.
- Helped contributors across several beginner-led programming-language projects through guidance, debugging, and occasional code contributions.

Selected Projects

- `datalog_par`: Standards-oriented Datalog compiler and parallel query engine written in Rust.
- `source_viewer`: Rust TUI for exploring disassembly and source mappings from DWARF debug information.
- `frame_mem_utils`: Stack-backed Rust arenas and vector-like data structures.

- `auto_new`: Lightweight Rust procedural macro for generating constructors without unnecessary stack copies.
- GPU Cellular Automata: Rust and wgpu simulation engine that runs cellular automata on the GPU.

Technical Skills

AI & ML: PyTorch, TensorFlow, Hugging Face, Diffusers, OpenVINO, Intel IPEX
LLMs & Tooling: LangChain, OpenAI API, FAISS
Systems: Rust, C, C++, x86-64 assembly, LLVM IR, CUDA (basic), Vulkan (basic), GLSL (basic), WGSL (basic), wgpu (basic)
Data & Scientific: NumPy, Pandas, scikit-learn, Matplotlib
Frontend & GUI (basic): JavaScript, HTML/CSS, egui, raylib, wgpu

Education

B.Sc. in Computer Science
- The Open University of Israel
- Currently in fourth year